Origin of Cells + Microscopy Study Guide

Section A2.2

A2.2.1— Cells as the basic structural unit of all living organisms. *NOS: Students should be aware that deductive reason can be used to generate predictions from theories. Based on cell theory, a newly discovered organism can be predicted to consist of one or more cells.*

- State the three parts of the cell theory.
    1. All living things are composed of cells
    2. The cell is the smallest unit of life
    3. All cells came from pre-existing cells
- Compare the use of the word theory in daily language and scientific language.

    Daily use: a guess

    Scientific: well-supported explanation based on extensive evidence

- Distinguish inductive from deductive reasoning.

    Inductive reasoning: going from the specific observation to the general

    Deductive reasoning: general to specific conclusion

- Outline the process of inductive reasoning that led to the development of the cell theory.

    Animal tissues have cells + plant tissues have cells, animal and plant tissues consist of cells, all organisms are made of cells

    Specific observations, pattern recognition, general conclusion

- Outline how deductive reasoning can be used to predict characteristics of a newly discovered organism.

    Existing theory, formulate hypothesis, collect data, analyze data, do/don't reject hypothesis

    If cell theory states all living organisms are made of cells, then a newly discovered organism predicted to be living should have one or more of cells

A2.2.2— Microscopy skills. *AOS: Students should have experience of making temporary mounts of cells and tissues, staining, measuring sizes using an eyepiece graticule, focusing with coarse and fine adjustments, calculating actual size and magnification, producing a scale bar and taking photographs. NOS: Students should appreciate that measurement using instruments is a form of quantitative observation.*

- Given the magnification of the ocular and objective lenses, calculate the total microscope magnification.

  Ocular always 10x * objective (4, 10, 40)

- Demonstrate how to focus the microscope on a sample.
- Demonstrate how to make a temporary wet mount and stain a microscopic sample.
- Demonstrate how to draw cell structures seen with a microscope using sharp, carefully joined lines and straight edge lines for labels.
- Measure the field of view diameter of a microscope under low power.

  Use ruler to measure

- Calculate the field of view diameter of a microscope under medium or high power.

  4x 4.50 → 4,500 micro

  10x 1.80 → 1,800 micro

  40x .45 → 450 micro

- Use a formula to calculate the magnification of a micrograph or drawing.

  M- magnification

  I- image size

  A- actual size

  m=i/a

- If given the magnification of a micrograph or drawing, use a formula to calculate the actual size of a specimen.

  a=i/m

- Compare quantitative and qualitative observations.

  Quantitative: numbers

  Qualitative: observations no numbers

A2.2.3— Developments in microscopy. *Include the advantages of electron microscopy, freeze fracture, cryogenic electron microscopy, and the use of fluorescent stains and immunofluorescence in light microscopy.*

- Define resolution and magnification.

  Resolution: smallest interval distinguishable by microscope (corresponds to the degree of detail visible in an image created by the instrument)

  Magnification: how much larger an object appears compared to its real size

- Compare the functionality of light and electron microscopes.

Light Microscope	Electron Microscope
Uses multiple lenses to bend light and magnify images	Uses electron beams focused by electromagnets to magnify and resolve
High ease of use	Low ease of use
Less expensive to buy	Expensive
Dead or living cells (if thin and unstained)	Dead cells
Yes movement	No movement
Yes colour	No colour
1500x max magnification (low)	100,000x to 300,000x magnification (high)
.25 micro to .30 micro (low resolution)	.001 micro (high resolution)

- State a benefit of using fluorescent stains to visualize cell structures.

  Generate bright images when chemicals in cells are monochrome (easier to see)

- Outline the process of visualizing specific proteins in cells using immunofluorescence technology.

  Antibodies bind to particular chemicals (antigens) in cell are produced, then fluorescent markers of diff colours are links to antibodies = multicoloured picture shown

- Outline the process of producing images of cell surfaces using freeze-fracture electron microscopy.
  1. Cell is frozen
  2. Cell is fractured at weakest points
  3. Cell is etched (surface of one side removes ice w evaporation)
  4. Vapor of platinum or carbon fired onto fracture to create metal coating
- Outline the process of visualizing specific proteins using cryogenic electron microscopy.
  1. Thin layer of protein
  2. Place thin layer of protein with solution (flash-frozen, to create smooth vitreous ice and prevent formation of water crystals, liquid ethane) on grid of electron microscope
  3. Record pattern or electrons transmitted by protein molecules
  4. Using patters, combine to produce 3D model of proteins

A2.2.4— Structures common to cells in all living organisms. *Typical cells have DNA as genetic material and a cytoplasm composed mainly of water, which is enclosed by a plasma membrane composed of lipids. Students should understand the reasons for these structures.*

- Outline the function of structures that are common to all cells.

  Plasma membrane: regulates what materials go in and out of cell

  Cytoplasm: gel-like fluid (mostly water) site of metabolic reactions

  DNA: genes containing information to make cell to carry out all of its functions

A2.2.5— Prokaryote cell structure. *Include these cell components: cell wall, plasma membrane, cytoplasm, naked DNA in a loop and 70S ribosomes. The type of prokaryotic cell structure required is that of Gram-positive eubacteria such as Bacillus and Staphylococcus. Students should appreciate that prokaryote cell structure varies. However, students are not required to know details of the variations such as the lack of cell walls in phytoplasmas and mycoplasmas.*

- Outline the functions of the following structures of an example prokaryotic cell: cell wall, plasma membrane, cytoplasm, 70s ribosome, and nucleoid DNA.

  Cell wall: provides structure and prevents the cell from bursting when it takes up water

  Plasma membrane: regulates what materials go in and out of cell

  Cytoplasm: gel like fluid (mostly h2o) site of metabolic reactions

  70s ribosome: catalyze synthesis of polypeptides, size 70 svedberg

  Nucleoid dna: not enclosed, single loop, not wrapped around proteins

- Define the term "naked" in relation to prokaryotic DNA.

  DNA is not wrapped around proteins

A2.2.7— Processes of life in unicellular organisms. *Include these functions: homeostasis, metabolism, nutrition, movement, excretion, growth, response to stimuli and reproduction.*

- List the common processes carried out by all life.

  Homeostasis: maintaining stable internal environment

  Metabolism: sum of all the chem reaction in a cell (enzyme catalyzed reactions)

  Nutrition: obtains energy

  Movement

Excretion: crating metabolic waste (co2 from breathing)

grow /develop

Response to stimuli: responds to environment

Reproduction: sexually or asexually, production of offspring

- Define metabolism, homeostasis, excretion, growth, nutrition, movement, reproduction and response to stimuli.

Homeostasis: maintaining internal body temp

Metabolism: sum of all the chem reaction in a cell (enzyme catalyzed reactions)

Nutrition: obtains energy

Movement

Excretion: crating metabolic waste (co2 from breathing)

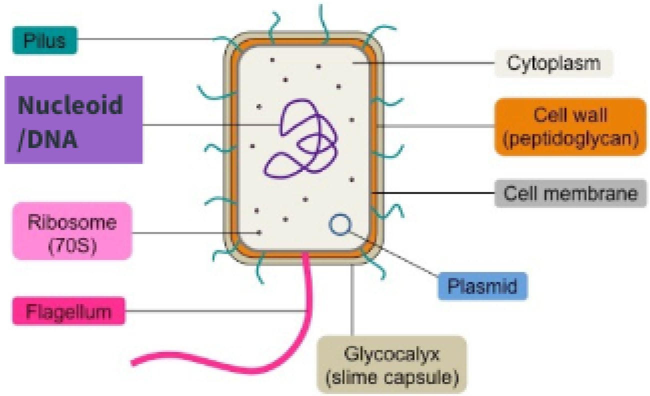
grow /develop

Response to stimuli: responds to environment

Reproduction: single or double

A2.2.10— Cell types and cell structures viewed in light and electron micrographs.
Application of skills: Students should be able to identify cells in light and electron micrographs as prokaryotes, ~~plant or animal.~~ In electron micrographs, students should be able to identify these structures: nucleoid region, prokaryotic cell wall, ~~nucleus, mitochondrion, chloroplast, sap vacuole, Golgi apparatus, rough and smooth endoplasmic reticulum, chromosomes,~~ ribosomes, cell wall, plasma membran~~e and microvilli~~ (plant and animal cells will be covered in a future unit)

- Recognize features and identify structures in micrographs of prokaryotic cells (inclusive of the plasma membrane, nucleoid region, ribosomes and cell wall).

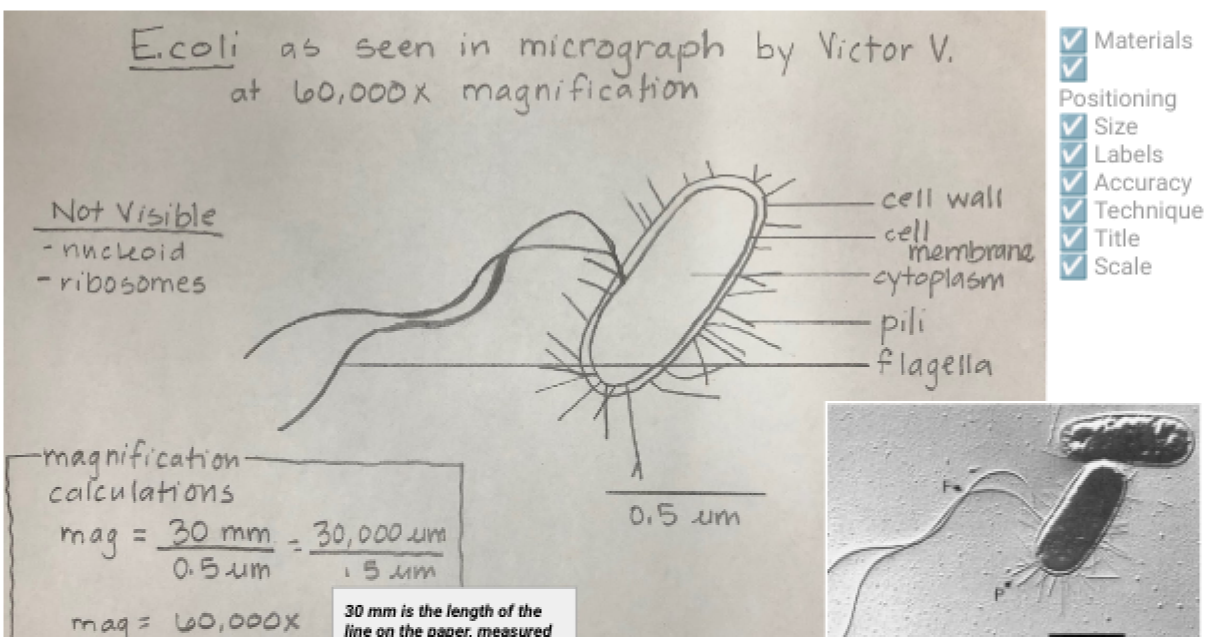


Key Features:

- *Pili* – shown as single lines
- *Flagella* – shown as thicker and significantly longer lines than the pili
- *Ribosomes* – labelled as 70S
- *Cell wall* – labelled as being composed of peptidoglycan; thicker than cell membrane
- *Shape* – appropriate for bacteria chosen (e.g. *E. coli* is a rod-shaped bacillus)
- *Size* – appropriate dimensions (e.g. length of cell twice the width)

A2.2.11— Drawing and annotation based on electron micrographs. *Application of skills: Students should be able to draw and annotate diagrams of ~~organelles (nucleus, mitochondria, chloroplasts, sap vacuole, Golgi apparatus, rough and smooth endoplasmic reticulum and chromosomes) as well as other~~ cell structures (cell wall, plasma membrane~~, secretory vesicles and microvilli~~) shown in electron micrographs. Students are required to include the functions in their annotations.* (plant and animal cells will be covered in a future unit)

- Given a micrograph, draw and label the ultrastructure of a prokaryotic cell.



<u>Section A2.1</u>

A2.1.1— Conditions on early Earth and the prebiotic formation of carbon compounds. *Include the lack of free oxygen and therefore ozone, higher concentrations of carbon dioxide and methane, resulting in higher temperatures and ultraviolet light penetration. The conditions may have caused a variety of carbon compounds to form spontaneously by chemical processes that do not now occur.*

- Outline the conditions that are thought to have existed on prebiotic Earth, including atmosphere, temperature, UV radiation, volcanic activity and asteroid bombardment.

  Atmosphere: "reducing atmosphere" high on reactive gasses (ammonia, hydrogen) little oxygen

  Temperature: very warm bc of astroids colliding

  UV radiation: no ozone layer means no block from UV, reaches surface

  Volcanic activity: eruptions lead to CO_2 and H_2O in atmosphere

  Asteroid bombardment: constant

- State that the conditions of prebiotic Earth may have caused a variety of carbon compounds to form spontaneously.

A2.1.2— Cells as the smallest units of self-sustaining life. *Discuss the differences between something that is living and something that is non-living. Include reasons that viruses are considered to be non-living.*

- Discuss the challenges of defining matter as living or nonliving.

  There are counter examples that don't meet requirements

- Discuss the reasons why cells are considered to be living.

  homeostasis, metabolism, nutrition, movement, growth n development, response to stimuli, reproduction (self replication singular)

- Discuss the reasons why viruses are considered to be non-living.

Not made of cells, don't grow, cant replicate themselves, cant perform independent metabolism

A2.1.3— Challenge of explaining the spontaneous origin of cells. *Cells are highly complex structures that can currently only be produced by division of pre-existing cells. Students should be aware that catalysis, self-replication of molecules, self-assembly and the emergence of compartmentalization were necessary requirements for the evolution of the first cells.*
NOS: *Students should appreciate that claims in science, including hypotheses and theories, must be testable. In some cases, scientists have to struggle with hypotheses that are difficult to test. In this case the exact conditions on pre-biotic Earth cannot be replicated and the first protocells did not fossilize.*

- Outline the intermediate stages needed for the evolution of the first cells on prebiotic Earth.
    1. Inorganic compounds to organic compounds
    2. Organic compounds to polymer
    3. Polymer to self replication
    4. Self replication to formation of cell
- Discuss limitations in testing hypotheses about the evolution of the first cells.

Conditions of earth were different

Not possible to replicate w/ certainty the conditions that would've existed

Well preserved fossils are rare

Methods used to estimate dates of 1st living cell have rangers of uncertainty

A2.1.4— Evidence for the origin of carbon compounds. *Evaluate the Miller–Urey experiment.*

- Outline the methodology, results and conclusion that can be drawn from Miller and Urey's experiments into the origin of biologically relevant carbon compounds.

  Methodology: water (representing ocean) heated, water evaporation travels to mix with ammonia, methane, and hydrogen gasses, electrodes to simulate lighting, cooling condenser turns steam into liquid for collection

  Results: carbon compounds, including amino acids, formed by inorganic

  Carbon compounds could spontaneously form on prebiotic earth

- Discuss the benefits and limitations of the Miller-Urey apparatus as a model for a natural phenomena.

  Benefits: Useful when direct observation or experimentation is difficult

  Limitations: Conditions of earth were different (atmosphere composition debated, lacked UV radiation, did not produce life only organic molecules)

Not possible to replicate w/ certainty the conditions that would've existed

Well preserved fossils are rare

Methods used to estimate dates of 1st living cell have ranges of uncertainty

A2.1.5— Spontaneous formation of vesicles by coalescence of fatty acids into spherical bilayers. *Formation of a membrane-bound compartment is needed to allow internal chemistry to become different from that outside the compartment.*

- Outline the cause and consequence of the spontaneous formation of membranes and vesicles by amphipathic molecules such as fatty acids and phospholipids on prebiotic Earth.

Fatty acids in a watery solution will be attracted to each other and will spontaneously form spherical structures called vesicles

Because polar & nonpolar exist, when in contact w water, will fold so polar outside nonpolar inside = movement of polar molecules into and out would've been limited by hydrophobic meaning developed own internal chemistry, diff from environment

A2.1.6— RNA as a presumed first genetic material. *RNA can be replicated and has some catalytic activity so it may have acted initially as both the genetic material and the enzymes of the earliest cells. Ribozymes in the ribosome are still used to catalyze peptide bond formation during protein synthesis.*

- State that modern cells use DNA as the genetic material and enzyme proteins as catalysts of metabolism.
- List properties of RNA that suggest it was the first genetic material.

-can store genetic info

-self replicating

-act as catalyst

-deoxyribose harder to make, in present day cells it's produced FROM ribose suggesting that ribose predates deoxyribose in cells

- Outline the ribosomal ribozyme as a type of RNA that is still used as a catalyst.

1. Binds to specific molecule (like substrate for enzymes)
2. Catalyzes a reaction that changes the substrate by breaking or building chemical bonds
3. Releases the product and is ready to do something to another molecule

-ribosome ribozyme can catalyze formation of peptide bonds during protein synthesis

A2.1.7— Evidence for a last universal common ancestor. *Include the universal genetic code and shared genes across all organisms. Include the likelihood of other forms of life having evolved but becoming extinct due to competition from the last universal common ancestor (LUCA) and descendants of LUCA.*

- Define LUCA.

Last universal common ancestor

- Discuss why the LUCA is not thought to be the first cell, but rather is thought to be the last common ancestor to all living cells.

    There was probably some other ancestor before LUCA (went extinct perchance), it's just that LUCA is the last universal one

- Explain the use of deductive reasoning to predict what genes were present in the LUCA cells.

    General: key parts in cells are ribosome and enzymes that synthesis DNA and RNA are all the same

    Specific: 350 occurring protein families identified as prokaryotes, that then trace back to LUCA

- List characteristics of the LUCA.

    -single cell prokaryotic (before nucleus)

    -has genetic info

    -contains universal genes shared among all life forms

    -carries out metabolism, replication, transcription, translation

A2.1.8— Approaches used to estimate dates of the first living cells and the last universal common ancestor. *Students should develop an appreciation of the immense length of time over which life has been evolving on Earth.*

- Compare the estimated dates for the evolution of the first cells and of the LUCA cells to the age of Earth.

    LUCA estimated before 3.9 b years ago, soon after formation of earth

    First cells around 3.8 to 4.0 b years ago not long after earth formed (4.54 b yrs ago)

    LUCA existed around 3.5 to 3.8 b ago

- Describe stromatolites as the earliest direct evidence of fossilized life.

Photosynthetic bacteria lived in shallow water, as they got covered in clay/particles, photosynthetic b move up to light, movement traps sediments into layers creating fossilized stromatolites

- Outline the use of isotopes and the molecular clock for estimating dates of the first cells and of the LUCA cells.

  Evidence of microscopic life trapped in rocks are analyzed using ratios of chemical isotopes

  Changes in mutation overtime, using molecular clock to count how many gene changes in sequences overtime

A2.1.9— Evidence for the evolution of the last universal common ancestor in the vicinity of hydrothermal vents. *Include fossilized evidence of life from ancient seafloor hydrothermal vent precipitates and evidence of conserved sequences from genomic analysis.*

- Explain the use of deductive reasoning to predict what genes were present in LUCA cells.

  General: genes needed for chemoautotrophic, anaerobic metabolism, were present in LUCA

  Environment w conditions for —-------- found in hydrothermal vents

  Specific: LUCA must have lived in the vicinity of hydrothermal vents

- Describe the conditions present at a white-smoker hydrothermal vent.

  Deep hydrothermal sea vents, gassy, metal rich, intensely hot plumes caused by seawater interacting with magma erupting through ocean floor

- Explain how knowledge of the genes present in the LUCA cells can provide evidence that the cells lived in the vicinity of hydrothermal vents.

  LUCA have genes adapted to environment:

    - Obligate anaerobic (no O2)
    - Chemoautotroph (creates food using chemical processes instead of photosynthesis)
    - Able to live in extreme heat

Section C3.1

C3.1.2— Cells, tissues, organs and body systems as a hierarchy of subsystems that are integrated in a multicellular living organism. *Students should appreciate that this integration is responsible for emergent properties. For example, a cheetah becomes an effective predator by integration of its body systems.*

- Define tissue, organ and organ systems.

Tissue: building blocks=cells
Organ: building block=tissues
Organ system: building block=organs

- Outline how integration occurs between and among tissues, organs and organ systems.
  Cell -> tissue -> organ -> organ system -> organisms
- Define emergent property.
  At each level of biological organization, new properties or behaviors emerge that are not present in the lower levels
  Part +part +part = whole —--> new characteristic
- State an example of an emergent property for each level of biological organization within a multicellular organism.
  Muscle cell —-> cardiac tissue —> heart organ —> vascular system —-> human

  Heart can pump blood but individual muscle cells cannot